

Title: A small molecule reacts with the p53 somatic mutant Y220C to rescue wild-type thermal stability

Running Title: A selective p53 Y220C covalent stabilizing compound

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Disclosure of Potential Conflicts of Interest:

Dr Shokat reports grants from The Emerald Foundation, grants from Samuel Waxman Cancer Research Foundation, and grants from Howard Hughes Medical Institute during the conduct of the study; personal fees from Apertor, personal fees from BioTheryX, personal fees from BridGene Biosciences, personal fees from Black Diamond, personal fees from Denali Therapeutics, personal fees from eFFECTOR Therapeutics, personal fees from Erasca, personal

fees from G Protein Therapeutics, personal fees from Genentech/Roche, personal fees from Ikena, personal fees from Initial Therapeutics, personal fees from Janssen Pharmaceuticals, personal fees from Kumquat Biosciences, personal fees from Kura Oncology, personal fees from Merck, personal fees from Mitokinin, personal fees from Nested, personal fees from Nextech, personal fees from Radd Pharma, personal fees from Revolution Medicines, personal fees from Rezo, personal fees from Totus, personal fees from Turning Point, personal fees from Type6 Therapeutics, personal fees from Wellspring Biosciences (Araxes Pharma), personal fees from Vevo, and personal fees from Vicinitas outside the submitted work; in addition, Dr Shokat has a patent for the KG series of molecules owned by UCSF pending and licensed to Nested; and I consult for Nextech which is a venture capital firm that has investments in PMV pharma which is developing p53 (Y220C) activating molecules. I do not consult for PMV pharma.

Dr Guiley reports personal fees from Nested Tx outside the submitted work; in addition, Dr Guiley has a patent for WO2021087096A1 issued and licensed to Nested Tx.

Abstract: The transcription factor and tumor suppressor protein p53 is the most frequently mutated and inactivated gene in cancer. Mutations in p53 result in deregulated cell proliferation and genomic instability, both hallmarks of cancer. There are currently no therapies available that directly target mutant p53 to rescue wild-type function. In this study we identify covalent compounds that selectively react with the p53 somatic mutant cysteine Y220C and restore wild-type thermal stability.

Statement of Significance: The tumor suppressor p53 is the most mutated gene in cancer and yet no therapeutics to date directly target the mutated protein to rescue wild-type function. In this study we identify the first allele-specific compound that selectively reacts with the cysteine p53 Y220C to rescue wild-type thermal stability and gene activation.

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Main Text:

Introduction

The most commonly mutated gene in cancer is the transcription factor and tumor suppressor *TP53* (p53). Unlike *RB1*, *CDKN2A*, and *PTEN*, which are lost in tumors through homozygous deletion, *TP53* is most frequently found with a somatic missense mutation in either a heterozygous setting or with a 17p deletion and loss of the second allele (1). A germline mutation in one *TP53* allele results in Li-Fraumeni syndrome, a disorder that increases the risk of cancer occurrence by 70-100% over an individual's lifetime (2). Cells carrying a *TP53* mutation accumulate high levels of mutant p53 protein, which drives a dominant-negative effect on the wild-type (WT) copy and the p53-homologs p63 and p73 (3). The primary transcriptional targets of p53 WT are p21 (*CDKN1A*), which functions as a potent cell cycle inhibitor, MDM2 (*MDM2*), the E3-ligase for p53, and pro-apoptotic Bcl-2 family proteins (*BBC3*, *BAX*, and *NOXA*). Genetically engineered mouse models have shown that restoration of p53 WT activity in p53-deficient cancers promotes tumor regression and a cure (4,5). The abundance of mutated p53 in tumors highlights the therapeutic potential of a small molecule capable of reverting mutant p53 to its WT form.

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The most frequent p53 missense mutations in cancer are found within the DNA-binding domain (DBD) (**Supplementary Fig. S1A and S1B**), where all mutations result in lower target

gene expression through the loss of DNA-binding affinity (6). Structural characterization of these hotspot mutations revealed a druggable cavity in p53 Y220C (7), a mutation that indirectly inhibits DNA binding through the loss of thermal stability in the DBD (**Supplementary Fig. S1C**). The small molecule PhiKan083 (8) was subsequently developed to bind within the p53 Y220C cavity and has undergone several chemical iterations to improve both affinity and thermal stabilization (9). Although the Phikan compounds have demonstrated the potential targeting of p53 Y220C, none have reached biochemical potency or showed thermal stabilization to WT levels, and thus are unlikely in their current form to satisfy the requirements of drug candidates.

The discovery (10) and recent FDA-approval of the cysteine-reactive K-Ras G12C inhibitor sotorasib (11) demonstrates the therapeutic potential of covalently targeting somatic mutant cysteines in proteins that lack canonical ligand-binding pockets. The weak reversible affinities of K-Ras G12C inhibitors were overcome by the kinetics of the acrylamide-cysteine Michael addition reaction to achieve high biochemical potency (12). In addition, several small molecule correctors of protein misfolding have also recently gained FDA-approval for treating cystic fibrosis (CFTR), Fabry disease (α GAL A), transthyretin amyloidosis (TTR), and sickle cell disease (HbS) (13).

In this study we identified the first covalent compounds that selectively react with the somatic cysteine in p53 Y220C to restore thermal stability to WT levels. Moreover, our structural analysis revealed a novel conformational state of the p53 DBD that serves as a potential druggable pocket for future studies. The findings and compounds presented herein represent a therapeutic strategy to treat over 125,000 cancer patients annually carrying the p53 Y220C mutation (9) and as a prophylactic therapy to lower the cancer risk in Li-Fraumeni syndrome.

Results

Covalent carbazoles labeled but minimally stabilized p53 Y220C

To generate a covalent p53 Y220C stabilizing compound we began with a carbazole scaffold and attached an acrylamide (KG1) or chloroacetamide (KG2) electrophile to the carbazole nitrogen (**Fig. 1A**). We hypothesized that these electrophiles would be positioned in the Y220C cavity similar to the ethyl group in PhiKan083 (8) (**Fig. 1A**) and would react directly with the mutant cysteine 220. To test Y220C-specific covalent labeling, we designed a cysteine light p53 Y220C DBD (Y220C-CL) (**Supplementary Fig. S1D**) by mutating the surface cysteines C124S, C182S, C229S, C275S and C277S. Notably, mild alkylating compounds have been found to react with C124, C182, and C277 (14,15), but not Y220C.

We incubated 10 μ M KG1 with 1 μ M p53 Y220C-CL at 4°C for 24 hr and found significant KG1 labeling ($92.9 \pm 0.4\%$) and no KG2 labeling (**Fig. 1B**). Cysteine light p53 WT (p53 WT-CL) showed no labeling for either compound indicating the reaction is specific to the Y220C cysteine. The inherent reactivity of chloroacetamide is greater than acrylamide towards thiols (16), suggesting that the acrylamide C β in KG1 is better positioned for the C220 nucleophilic attack compared with the C α in the chloroacetamide KG2 (**Fig. 1A**).

To determine the stabilizing effect of KG1, we first measured the melting temperature (T_m) for p53 Y220C-CL (33.21 ± 0.02 °C) and p53 WT-CL (41.59 ± 0.02 °C) alone and calculated the change in T_m (ΔT_m) for the Y220C mutation to be -8.40 ± 0.02 °C. (**Supplementary Fig. S2A**). We then fully labeled p53 Y220C-CL with KG1 and purified the complex by gel filtration to remove excess compound. The ΔT_m for KG1 was determined to be $+1.28 \pm 0.01$ °C, similar to the PhiKan083 ΔT_m of $+1.11 \pm 0.06$ °C at 250 μ M (**Supplementary Fig. S2A**).

In an effort to improve the covalent carbazole interaction with p53 Y220C, we installed H-bond donating groups at the 3-position (**Fig. 1A**) to establish a H-bond interaction with the D228 backbone carbonyl (8). The amide KG5 provided the greatest labeling ($95.2 \pm 0.5\%$) compared to the aldehyde KG3 ($79.0 \pm 1.0\%$) or methylamine KG4 ($84 \pm 1\%$) (**Fig. 1B; Supplementary Fig. S2A, S2B, S2C, and S2D**). To determine whether KG5 engages endogenous p53 Y220C, we treated HepG2 (p53 +/+) or Huh-7 (p53 Y220C/-) cells with 5-25 μ M KG5 for 1 hr and performed a click reaction between TAMRA-azide and KG5's alkyne in cell lysate to induce a mobility shift visible by SDS-PAGE western blot. KG5 was found to fully engage p53 Y220C at 20 μ M with no engagement observed in WT (**Fig. 1C**).

To determine the binding mode of the covalent carbazole, we solved a co-crystal structure of p53 Y220C-CL bound to KG3 at 2.4 Å resolution (**Fig. 1D; Supplementary Table S1**). The overall fold of the p53 Y220C-CL is comparable to p53 Y220C-PhiKan083, with a root-mean-square deviation (RMSD) of 0.383 Å; however, there are several differences in the small molecule's binding mode. The 3-positions of the two carbazoles face opposing sides of the p53 Y220C cavity, where the aldehyde on KG3 forms a H-bond with T150, whereas the 3-position methylamine on PhiKan083 forms a H-bond with the carbonyl backbone of D228 (**Fig. 1D and 1E**). The covalent KG3-C220 bond orients the acrylamide C α and C β outside of the Y220C hydrophobic cavity between C220 and L145, resulting in the loss of the van der Waals (vdW) interactions with p53 V147, L145, F109, L257, and V157, which were observed with the ethyl group in PhiKan083 (**Fig. 1E**) or pyrrole in other reversible compounds (17). Interestingly, it was originally predicted through computational docking that PhiKan083 would adopt the same conformation that we observed for KG3 (8).

While our covalent carbazole series labeled recombinant and endogenous p53 Y220C, the compounds provided only minimal thermal stabilization. The p53 Y220C-KG3 crystal structure presented an opportunity to enhance stability and labeling by creating a new series of compounds. We hypothesized that positioning a methyl group on the C α of the acrylamide would mimic the ethyl group in Phikan083 and gain vdW interactions with p53 V147, L145, F109, L257, and V157. In addition, the altered binding conformation of the carbazole ring suggested that installing H-bond donors on the 4-position would establish interactions with the D228 carbonyl (**Fig. 1D**).

Methacrylamide indole series enhanced stabilization of p53 Y220C and induced a novel conformation

Insertion of a C α methyl acrylamide onto the carbazole produced a labile amide bond to the carbazole presumably due to a steric clash with the phenyl ring of the carbazole. To avoid this clash, the carbazole scaffold was replaced with an indole (**Fig. 2A**). We found that the indole methacrylamide KG6 did not react as efficiently as KG3, with $15 \pm 1\%$ labeling compared with $79 \pm 1\%$, respectively (**Fig. 2B; Supplementary Fig. S3A, S3B, S3C, S3D, and S3E**). However, the thermal stability was enhanced with a ΔT_m of $+3.563 \pm 0.002$ °C for KG6 compared with $+1.4 \pm 0.1$ °C for KG3. Indeed, a C α methyl substitution on an acrylamide is known to substantially lower the reactivity towards thiols due to steric hindrance of the nucleophile (18). We next tested substitutions at the 4-position of the indole to enhance labeling efficiency through ionic interactions with the p53 D228 carbonyl (**Fig. 2A**). The phenyl-piperazine compound KG10 was found to have improved labeling ($51.6 \pm 0.7\%$), while maintaining a significant shift in thermal stability ($\Delta T_m +3.21 \pm 0.03$ °C).

To determine the structural mechanism of stabilization by the methacrylamide indole series, we solved co-crystal structures of p53 Y220C-CL in complex with KG6 or KG10 at 1.6 Å

and 2.0 Å, respectively (**Supplementary Table S1 and Fig. S4A, S4B, S4C, and S4D**).

Strikingly, the p53 Y220C-KG6 complex shows significant structural changes in L6 where the C α position of E224 is altered by 13.2 Å (**Fig. 2C**) relative to p53 Y220C-CL-KG3 while the overall fold remains similar (RMSD of 0.454 Å). This closed conformation of L6 in the p53 Y220C-KG6 complex has not been observed in previous p53 crystal structures. The significant stability enhancement by KG6 is driven by establishing a hydrophobic network within the core of the immunoglobulin fold at β -strands S3, S5, S7, and S8 (**Fig. 2C**). vdW contacts are formed between the C β of the acrylamide on KG6 with p53 V157 and V218, and between the C α methyl group of the acrylamide on KG6 with L145, V218, T230, and I232. The KG6 methyl acrylamide contacts are bridged to β -strands S3 and S8 through interactions between the indole ring and p53 T230 and V147. The alternate binding mode of KG6 compared to KG3 positions the aldehyde of KG6 to form a H-bond interaction with the backbone amine of S227 instead of T150. Finally, an allosteric stabilization effect was observed as result of the L6 movement, where the D228 carbonyl is shifted 2.9 Å, establishing an intramolecular H-bond with the backbone amine of V147 rather than through the small molecule itself (**Supplementary Fig. S4E**).

While KG6 induces a novel p53 Y220C conformation, the L6 position and overall fold in the Y220C-KG10 complex is similar to KG3, with an RMSD of 0.367 Å (**Fig. 2D**). The elongated KG10 molecule makes distal contacts with p53 Y220C spanning from the core of the immunoglobulin fold to L6 D228 outside of the pocket (**Fig. 2D**). As predicted, the C α methyl of the acrylamide is positioned in the C220-L145 cavity similar to the PhiKan083 ethyl (**Fig. 1E and 2D**), making vdW interactions with p53 F109, L145, V147, and L257. The piperazine makes H-bond interactions with D228, and a D228A mutation reduced labeling from 51.6 ± 0.7 % to 37.2 ± 0.5 %. No change in labeling was observed with the D228A mutation from compounds KG6 and

KG7, which lack the piperazine moiety. The labeling of compound KG7 was not equivalent to KG10 with the D228A mutation, suggesting that the piperazine is also making favorable vdW contacts with p53 W146. In contrast to KG3, the KG10 3-position aldehyde is oriented into the Y220C cavity and forms a H-bond with the backbone amine of T230. However, W146 is shifted 0.8 Å away from the cavity relative to Y220C-KG3 and 1.5 Å relative to Y220C-KG6.

Methacrylamide 6-azaindole rescued WT DBD stability

While we predicted the aldehyde in KG10 would interact with T150, the crystal structure revealed that the 3-position was facing the hydrophobic pocket formed by V147, W146, and C229 on the opposing side of the Y220C crevice (**Fig. 2D**). A methyl substitution at the 3-position position would be predicted to favor the hydrophobic pocket and removal of the electron-withdrawing group from the ring would reduce the electrophilicity of the acrylamide thereby lowering off-target engagement in cells. In addition, p53 T150 is positioned adjacent to C6 of the indole ring in the Y220C-KG10 structure (**Fig. 2D**) suggesting that a nitrogen substitution at C6 would further enhance stability and labeling through a H-bond interaction with T150. Strikingly, we found that these substitutions together enhanced the thermal stability of p53 Y220C near that of WT, with a ΔT_m of $+7.8 \pm 0.1$ °C (**Fig. 3A and 3B**).

Having observed a significant thermal shift after changing the core scaffold from an indole to an azaindole, we next tested the thermal stability contribution of the D228 interaction with phenyl-piperazine in KG11. The D228A mutation showed no difference in overall p53 thermal stability in the absence of ligand, with a T_m of 33.47 ± 0.2 °C for p53 Y220C-CL D228A compared with 33.21 ± 0.02 °C for p53 Y220C-CL. However, the ΔT_m when bound to KG11 was $+5.91 \pm 0.05$ °C for p53 Y220C-CL D228A compared with $+7.8 \pm 0.1$ °C for p53 Y220C-CL, highlighting the significance of this ionic interaction for both labeling efficiency and

stabilization (**Fig. 3B**). To further strengthen the D228 interaction, we substituted the piperazine for a stronger base, piperidine. The substitution resulted in increased labeling from 29 ± 1 % for KG11 to 53.4 ± 0.3 % for KG12. Finally, to make the molecule more rigid, the phenyl piperidine (KG12) was fused to produce a tetrahydroisoquinoline (KG13). The KG13 azaindole molecule labeled p53 Y220C-CL at 65 ± 2 % and stabilized p53 Y220C back to WT levels with a ΔT_m of $+8.3 \pm 0.1$ °C (**Fig. 3B**).

To determine the structural mechanism of stabilization by the azaindole KG13, we solved a co-crystal structure of p53 Y220C-CL in complex with KG13 at 1.7 Å (**Fig. 3C**; **Supplementary Table S1**). The binding mode and overall protein fold is similar to p53 Y220C-KG10, with a RMSD of 0.095 Å. As predicted, the aldehyde to methyl substitution at the 3-position resulted in W146 orienting 2.7 Å back into the Y220C crevice, being positioned 3.5 Å from the methyl group and making vdW contacts. The N6 substitution to the indole ring shifted the ring 0.6 Å towards the hydroxyl in T150, and established a new H-bond interaction (**Fig. 3C**). The Cα of D228A is shifted 1.8 Å relative to p53 Y220C-KG10 with the fused ring piperidine positioned to establish a H-bond with D228. A D228A mutation resulted in reduced KG13 labeling efficiency from 65 ± 2 % for p53 Y220C-CL to 50 ± 1 % for p53 Y220C-CL D228A (**Fig. 3D**).

We next tested whether the azaindole KG13 alters DNA binding specificity. Using an electrophoretic mobility shift assay (EMSA), we tested *CDKN1A* promoter binding with p53 WT-CL, p53 Y220C-CL, and p53 Y220C-CL-KG13. We found the apparent K_d for *CDKN1A* to be between 2-3 μM for all p53 WT and Y220C samples tested at 25°C, suggesting DNA specificity is not compromised following KG13 binding to p53 Y220C (**Supplementary Fig. S5A, S5B**,

S5C, S5D, and S5E). In contrast, the DNA-contact mutant R273C-CL showed negligible binding (**Supplementary Fig. S5F**).

Azaindole series specifically labeled p53 Y220C but not WT cysteines

Our goal of generating a mutant-specific p53 Y220C reactivating compound required selectivity towards the somatic mutant cysteine Y220C. Sulfonylpyrimidines have been shown to alkylate the p53 DBD non-specifically, where PK11000 reacts with cysteines 182 and 277 (15). Using PK11000 as a control non-specific alkylating compound, we tested 10 μ M PK11000 for 24 hr at 4°C on a non-cysteine light p53 DBD with (p53 Y220C) or without (p53 WT) the Y220C mutation and observed two labeling events on both proteins (**Fig. 4A**). We next tested the carbazole KG5, indole scaffolds KG6 or KG7, and the azaindole series KG11, KG12, and KG13 (**Fig. 4A**) on p53 WT and Y220C. The carbazole KG5 was found to be highly reactive with several labeling events on both p53 WT and Y220C. The indole scaffolds KG6 and KG7 also labeled both p53 WT and p53 Y220C; however, the total labeling events were reduced to two, with greater selectivity for p53 Y220C compared with PK11000 (**Fig. 4A**). The difference in reactivity between the carbazole KG5 and the indoles KG6 or KG7 is likely attributed to reducing the electrophilicity of the acrylamide through the C α methyl substitution. Finally, KG11, KG12, and KG13 labeled 43.2 ± 0.4 %, 68.7 ± 0.7 %, and 78 ± 2 % respectively for p53 Y220C with no labeling of WT (**Fig. 4A**).

Azaindole stabilized cellular p53 Y220C and activated p53 target genes

To determine whether a Y220C-specific covalent compound could rescue p53 Y220C stability in cells we performed a cellular thermal shift assay (CETSA) using KG13. H1299 cells (p53 -/-) stably expressing p53 Y220C were treated with DMSO or 100 μ M KG13 for 1 hr, harvested, and heat treated from 30-50°C before being analyzed by western blot. Given that p53

Y220C mutant is unstable at 37°C, we incubated the cells at 28°C for 24 hrs prior to and during compound treatment to obtain the melting curve. The WT-specific conformational antibody Pab1620 (19) was able to immunoprecipitate p53 Y220C at 28°C but not 37°C, confirming proper folding of the DBD at the permissive temperature (**Supplementary Fig. S6A, S6B, and S6C**). In the absence of stabilizing ligand, p53 Y220C began aggregating at 37°C with an aggregation temperature (Tagg) of 39 ± 1 °C (**Fig. 4B**). For p53 WT, we observed no difference in protein stability between 30-37°C with a Tagg 44 ± 1 °C, consistent with full-length recombinant p53 WT (20). Finally, we treated cells expressing p53 Y220C with 100 μM KG13 for 1 hr and observed a rescue of stability at 37°C, and a shift in Tagg to 43 ± 1 °C, similar to p53 WT (**Fig. 4B; Supplementary Fig. S6D, S6E, and S6F**).

After observing KG13 stabilized cellular p53 Y220C, we next monitored the compound's effect on gene expression, protein induction, and p53 Y220C promoter occupancy upon warming cells to physiological temperature. The isogenic cell panels H1299 (p53 -/-) and U2OS (p53 +/+) stably expressing p53 Y220C were treated with 25 μM KG13 for 1 hr at 28°C, washed with media, and transferred to 37°C for 8 hrs before analysis. *CDKN1A* was the most upregulated p53 target gene by KG13, with a 7 ± 1 -fold and 10.6 ± 0.4 -fold increase in expression in H1299 and U2OS p53 Y220C expressing cells respectively (**Fig. 4C and 4D; Supplementary Fig. S7A**). Moreover, KG13 was found to prolong p53 Y220C occupancy on *CDKN1A* and *MDM2* promoters in H1299 and U2OS cells (**Supplementary Fig. S7B and S7C**). To compare KG13 mediated reactivation of p53 Y220C to p53 WT activity, we treated the U2OS cell panel with the MDM2 inhibitor Nutlin-3a and observed a 8.5 ± 0.2 -fold increase in *CDKN1A* in the parental cells (**Fig. 4D**). The p53 Y220C expressing U2OS cells also responded Nutlin-3a, however the effect was suppressed due to the dominant-negative effect on p53 WT (**Fig. 4D**).

To test the therapeutic potential of KG13, we treated the patient derived cells NUGC-4 (p53 +/+), NUGC-3 (p53 Y220C/+) and BxPC-3 (p53 Y220C/-) with KG13, and monitored target gene expression, growth inhibition, and caspase activity. We first performed a time-course western blot after treating the cells with DMSO, 10 μ M Nutlin-3a or KG13 at 28°C, and monitored p53, p21, and MDM2 protein levels upon returning to 37°C. In DMSO treated cells both p53 Y220C cell lines NUGC-3 and BxPC-3 show a potent induction of MDM2 and p21 at 28°C that dissipates over the 8 hr shift to 37°C (**Fig. 4E**). p53 WT also shows greater activity in NUGC-4 cells at 28°C, however the induction of MDM2 and p21 is less pronounced compared to the temperature-sensitive mutants, consistent to what has been observed previously (21). In the Nutlin-3a treated cell panel, only p53 WT in NUGC-4 shows potent induction of MDM2 and p21 starting at 4 hrs post shift to 37°C (**Fig. 4E**). Notably, Nutlin-3a does not stimulate p53 Y220C activity even when the protein is functional at 28°C, suggesting an alternative degradation pathway for the temperature-sensitive mutant. In the KG13 treated panel we observed elevated MDM2 and p21 protein levels in NUGC-3 and BxPC-3 compared to DMSO or Nutlin-3a treated cells, whereas no effect was observed in the p53 WT NUGC-4 cells. In the patient-derived p53 Y220C cell panel, *PUMA* was the most upregulated gene in response to 25 μ M KG13, with an 5.1 ± 0.2 -fold and 5.3 ± 0.8 -fold increase in NUGC-3 and BxPC-3 cells respectively (**Supplementary Fig. S7D**). Finally, we monitored growth inhibition and caspase 3/7 activity following KG13 treatment on the patient-derived cell lines and found an IC_{50} of 14.7 ± 0.6 μ M, 7.1 ± 0.2 μ M, and 11.8 ± 0.6 μ M for NUGC-4, NUGC-3, and BxPC-3 respectively with an increase in caspase 3/7 activity only observed in p53 Y220C cells (**Fig. 4F; Supplementary Fig. S7E and S7F**). In summary, KG13 treated cells displayed p53 Y220C-dependent activation of p53 target genes with growth inhibition and increased caspase activity.

Discussion

In this study, we have identified the first Y220C-specific covalent compounds that enhance mutant p53 thermal stability to WT levels. Our structural analysis revealed two conformational states of the covalently modified p53 Y220C, with the KG6-Y220C L6 state representing a novel druggable pocket. Previous Markov state models and NMR studies have proposed dynamic conformations of L6, suggesting this pocket also exists in an unliganded state (22). While the p53 D228 backbone carbonyl represents an important docking site for the PhiKan reversible compounds (8,9), we found that the covalent molecules presented here rely on a direct interaction with the D228 carboxylate for thermal stability and labeling efficiency (**Fig. 2D and 3C**).

Similar to the development of K-Ras G12C inhibitors, the reactivity of KG13 towards the p53 somatic mutant cysteine Y220C provides a precision medicine approach to generate p53 WT activity specifically in tumor cells harboring the p53 Y220C mutation (**Fig. 4C, 4D, and 4F**). This approach would circumvent the toxicity observed through pan-p53 activation with MDM2 inhibitor compounds (23). Moreover, we observed that Nutlin-3a does not stimulate p53 Y220C activity at permissive temperatures when the mutant is folded in a WT like state and activating target gene expression (**Fig. 4E; Supplementary Fig. S7**). The KG13 molecular chaperone mechanism of rescuing the thermal stability of the p53 DBD fold has precedent in the clinic through voxelotor, which also covalently reacts with a tetrameric target to prevent cellular aggregation (24). Future studies will be required to determine how to achieve the maximum tumor suppressive effect from p53 Y220C stabilizing compounds, which could be achieved through combination with cell-cycle inhibitors, pro-apoptotic drugs, or through therapeutic hypothermia (21).

Materials and Methods:

Recombinant protein expression and purification

Human p53 WT, Y220C, and R273C “cys-light” DBD (residues 94-312, (C124S, M133L, C182S, C229S, N239Y, N268D, C275S, C277S) p53 WT-CL, p53 Y220C-CL, p53 R273C-CL) or “non-cysteine light” (residues 94-312, (M133L, V203A, N239Y, N268D) p53 WT, p53 Y220C) were expressed as 6X HIS fusion proteins in *E. coli* BL21(DE3). Bacterial cultures were grown to 0.4 OD₆₀₀ and induced with 1 mM IPTG (GoldBio) and 200 μM ZnSO₄ at 18°C for 16 hrs. Cells were lysed by sonication in Lysis Buffer 50 mM HEPES, 500 mM NaCl, 5% glycerol, and 20 mM Imidazole (pH 7.0), clarified by centrifugation, and loaded onto Ni Sepharose High Performance (GE Healthcare) affinity resin. Following 10 CV of washing with Lysis Buffer, proteins were eluted from the resin in HIS Elution Buffer 50 mM HEPES and 400 mM Imidazole (pH 7) and diluted in Buffer HA 50 mM HEPES, 20 mM NaCl, and 15% glycerol (pH 7.0) such that the solution <100 mM NaCl. The protein was then loaded onto HiTrap Heparin HP (Cytiva) affinity chromatography and eluted using Buffer HA and Buffer HB 50 mM HEPES, 1 M NaCl, and 15% glycerol (pH 7.0) with a gradient 0-100% Buffer HB over 100 mL. The Heparin elution fraction was then subjected to HIS-TEV protease cleavage overnight in 50 mM HEPES, 200 mM NaCl, and 5 mM DTT (pH 7.0). The protein was then passed over HiTrap Heparin HP (Cytiva) affinity resin again to remove free 6XHIS and HIS-TEV protease and was flash frozen in liquid nitrogen.

Crystallization, data collection, structure determination, and model refinement

p53 Y220C-CL was prepared for crystallization by treatment with excess covalent compound in a buffer containing 50 mM HEPES and 150 mM NaCl pH 7.0 for 24 hrs at 4°C. The p53 Y220C-

CL-small molecule complex was then eluted from a Superdex 75 column (GE Healthcare) in a buffer containing 10 mM Tris and 100 mM NaCl pH 8.0. The p53 Y220C-CL-small molecule complex was crystallized from a 10 mg/mL solution by hanging drop vapor diffusion method at 22°C. Pyramidal crystals for KG3 and KG6, or rod shaped crystals for KG10 and KG13 formed after 2-6 days in 100 mM HEPES, 2.2M MgSO₄ pH 7.0. Crystals were cryo-protected in reservoir solution supplemented with 25% ethylene glycol and cryo-cooled in liquid nitrogen.

Data were collected at the Advanced Light Source, Laurence Berkeley National Laboratory at beamline 8.2.1. Diffraction spots were integrated using MOSFLM, and data were merged and scaled using Scala in the CCP4 software package. The model was built with Coot, and the model was refined with Phenix (25). Phenix Xtriage indicated a perfect merohedral twin for the KG3 dataset and the -h,-k,l twin law was applied to refinement.

Cell lines

H1299 (ATCC, CRL-5803), HUH-7 (UCSF, Jura Lab), U2OS (UCSF, Cell culture facility), NUGC-3 (JCRB, JCRB0822), NUGC-4 (JCRB, JCRB0834), BxPC-3 (ATCC, CRL-1687), and HepG2 (UCSF, Cell culture facility). ATCC and JCRB cell lines were tested for *Mycoplasma* prior to arrival. All UCSF Cell culture facility lines were tested for endotoxin and STR validated by the facility. HUH-7 cells were not tested.

Western blots, Immunoprecipitation, and antibodies

Whole-cell extracts were prepared by lysing cells with lysis buffer containing 50 mM HEPES, 150 mM NaCl, 1 mM DTT, and 0.1% NP-40 (pH 7.0) in the presence of 1X cOmplete EDTA-free protease inhibitor cocktail (Roche). Whole-cell extracts were combined with 2X SDS-loading buffer for western blot analysis.

Western blots were performed with 5 µg total protein lysate and were resolved by SDS-PAGE on 4-12% BT gels (Invitrogen) at 150 V for 1 hr in MES buffer and transferred to nitrocellulose membranes, which were then incubated with primary antibodies at 4 °C overnight, followed by incubation with LI-COR IRDye anti-mouse and anti-rabbit secondary antibodies at room temperature for 1 hr. Bands were imaged on a LI-COR Odyssey scanner.

For immunoprecipitation, cell extracts from H1299 cells were prepared by sonicating in an IP Lysis Buffer containing 50 mM Tris, 150 mM NaCl, 1 mM DTT, 10% glycerol, and 0.1% Tween-20 (pH 8) in the presence of 1X Sigma protease inhibitor cocktail. 400 µg of total protein from cleared lysate was then incubated with 2 µg Pab1620 (MABE339, Millipore) for 4 hrs at 4°C with 20 µL protein A/G beads in 500 µL IP Lysis Buffer. The mixture was then washed 3X with IP Lysis Buffer, and eluted with 2X SDS-loading buffer for western blot analysis.

For compound treatment (**Fig. 4E**), the same protocol was followed as RT-qPCR.

Antibodies for western blot detection were as follows: p53 (DO-1, SCBT), p53 (7F5, CST), MDM2 (D1V2Z, CST), β-actin (8H10D10, CST), and p21 (SX118, SCBT).

LC-MS analysis of p53 Y220C covalent labeling

Recombinant p53 (1 µM) in a buffer containing 50 mM HEPES and 150 mM NaCl (pH 7.0) was treated with covalent compounds (10 µM) for 24 hrs at 4°C. The reactions were quenched by addition of 10 mM DTT. The extent of covalent labeling was assessed by LC-MS (Waters Xevo G2-XS QToF, ACQUITY UPLC Protein BEH C4 Column, 300Å, 1.7 µm, 2.1 mm X 50 mm). Deconvolution of multiply charged ions was performed using Waters MassLynx software (version 4.1). Reactions were performed in duplicate.

Differential Scanning Fluorimetry

Recombinant p53 Y220C-CL was treated with excess covalent compound in buffer containing 50 mM HEPES and 150 mM NaCl (pH 7.0) for 24 hrs at 4°C. The p53 Y220-CL-small molecule complex was then eluted from a Superdex 75 column (GE Healthcare) in a buffer containing 50 mM HEPES and 150 mM NaCl (pH 7.0) and the peak fraction was isolated. 5 μ M of the p53 Y220C-CL-small molecule complex, 2X SYPRO Orange (Invitrogen), 50 mM HEPES, and 150 mM NaCl (pH 7.0) were mixed in a 96-well white PCR plate (USA Scientific) at a volume of 25 μ L per well. The plate was sealed with optically clear PCR sealing film (USA Scientific). The thermal shift assay was performed on a Bio-Rad CFX96 qPCR system. The temperature was increased from 15°C to 70°C at a rate of 0.5°C/s. After each temperature step, fluorescence was monitored with an excitation of 492 nm and emission of 610 nm. Each sample was run in triplicate. “cys-light” WT was analyzed under the same conditions without compound. T_m was calculated using the Boltzmann function (Graphpad Prism 8).

RT-qPCR

0.3×10^6 cells were plated in 6-well plates and allowed to recover for 24 hr at 37°C. Cells were transferred to 28°C for 16 hrs before being treated with KG13 for 1 hr. The compound was washed out with DMEM, and the cells recovered at 37°C for 8 hrs. Nutlin-3a and DMSO treated samples were also incubated at 28°C for 1 hr, then left in the media during the 8 hr recovery at 37°C. RNA was isolated from cells using RNeasy Plus Mini Kit (QIAGEN # 74004) and SuperScript III First-Strand Synthesis SuperMix for qRT-PCR (Life Technologies # 11752050) was used for reverse transcription reaction. qPCR reactions were then performed using the

Maxima SYBR Green qPCR Master Mix (Life Technologies # K0221) on a Bio-Rad CFX96 qPCR system system. *GAPDH* served as the reference gene. All data was evaluated using the $\Delta\Delta C_t$ method. Primer sequences: *GAPDH* forward: 5'- GACCCCTTCATTGACCTCAAC-3', *GAPDH* reverse: 5'- CACGACGTACTCAGCGCC-3'; *CDKN1A* forward: 5'- GGAAGACCATGTGGACCTGT-3', *CDKN1A* reverse: 5'-GGATTAGGGCTTCCTCTTGG-3'; *MDM2* forward: 5'-GTGAATCTACAGGGACGCCA-3', *MDM2* reverse: 5'-CTGATCCAACCAATCACCTGAA-3'; *BBC3* forward: 5'-CCTGGAGGGTCCTGTACAATCT-3', *BBC3* reverse: 5'-TCTGTGGCCCCTGGGTAAG-3'. Samples were run in technical triplicate and biological duplicate.

Cellular Thermal Shift Assay (CETSA)

To generate p53 Y220C expressing cells, H1299 cells (p53 $-/-$) were transfected with Y220C modified p53 plasmid (addgene #49242) using FuGENE 6 (Promega #E2691). Stable cells were selected with G418 for 14 days. 3×10^6 cells were plated in 10 cm dishes and were allowed to recover overnight at 37°C. The cells were then transferred to a 28°C incubator for 24 hrs, then treated with DMSO or KG13 for 1 hr. Cells were harvested by scraping in PBS, and pelleted at 3500 rpm for 5 minutes. Cells were washed in PBS and spun down again for 5 minutes. Cells were resuspended in 450 uL PBS with 1X cOmplete EDTA-free protease inhibitor cocktail (Sigma # 11836170001) and split into 50 uL aliquots in an 8-tube strip (USA Scientific #1402-4700). The cell suspension was heated with a gradient from 30-50 °C in a thermocycler for 3 minutes, then cooled to 25 °C for 3 minutes. The heat-treated samples were flash frozen in liquid nitrogen, thawed at 25°C and briefly vortexed. The freeze-thaw cycle was repeated three times total. Samples were cleared of aggregate by centrifugation at 20,000 g and 4°C for 20 minutes.

uL of cleared supernatant was mixed with 5X SDS-loading buffer for western blot analysis. For WT p53, the same protocol was followed except H1299 cells were transiently transfected. The intensity ratio from anti-p53 rabbit (CST #7F5) to anti-actin mouse (CST #8H10D10) was calculated and normalized to the starting point at 30°C. Each sample was run in biological duplicate and Tagg was calculated using the Boltzmann function (Graphpad Prism 8).

Electrophoretic Mobility Shift Assay (EMSA)

The p53 binding site of the *CDKN1A* promoter was assembled by heating an equimolar mixture of oligonucleotides forward- 5'-AAAGGAACATGTCCCAACATGTTGAGAA-3' and reverse- 5'-TTCTCAACATGTTGGGACATGTTTCCTTT-3' to 95 °C for 5 minutes, and cooling to 25°C over 1 hr. 2 µM *CDKN1A* was incubated with a p53 DBD gradient ranging from 0-20 µM at 25°C for 30 minutes in 0.5 X TBE buffer. Following incubation, samples were mixed 1:1 with 2X Native loading buffer (62.5 mM Tris pH 7, 40% glycerol, 0.01% bromophenol blue), loaded onto a DNA retardation gel (Thermo Fisher #EC63655BOX) and run at 75V for 1 hr at 25°C. The gel was stained in 1 X SYBR Safe DNA Gel stain (Thermo #S33102) with 0.5 X TBE buffer (Thermo Fisher #S33102) for 10 minutes before being analyzed by a blue light transilluminator. The top intensities of the gel shift were calculated using ImageJ, and the Kd values were calculated using specific binding function (Graphpad Prism 8).

Cell Viability and Caspase 3/7 Assay

Cells were seeded into 96-well plates (Greiner Bio-One #655083) at 3×10^3 cells per well and were allowed to recover at 37°C overnight. Cells were then transferred to 28°C for 16 hr before being treated with the indicated compound for 1 hr. KG13 was then washed out with DMEM, and

the cells were transferred to 37°C for 48 hr. Cells treated with Nutlin-3a were transferred to 37°C for 48 hr without the washout. CellTiter-Glo (Promega #G7572) luminescence-based assay was diluted 1:5 in PBS and was mixed directly in the cell wells as a 1:1 mixture. The reaction mixture was incubated at 30 minutes while shaking at 25°C before being analyzed on a Tecan Spark plate reader. For caspase 3/7 activity, Caspse-Glo 3/7 (Promega #G8091) was added to the cells at a 1:1 mixture 24 hr following compound treatment. The mixture was incubated at 25°C for 10 minutes while shaking before being analyzed on a Tecan Spark plate reader.

ChIP Assay

Cells were seeded into 150 mm-dishes at 70% confluency and were allowed to recover at 37°C overnight. Cells were then transferred to 28°C for 16 hr before being treated with the indicated compound for 1 hr. KG13 was then washed out with DMEM, and the cells were transferred to 37°C for 1 or 2 hrs. Cells treated with DMSO or Nutlin-3a were transferred to 37°C for 1 or 2 hrs without the washout. Cells were harvested and cross-linked in 1% formaldehyde in PBS for 10 minutes at room temperature. The Zymo-Spin ChIP Kit (Zymo Research # D5210) was used for DNA immunoprecipitation and purification except agarose beads (SCBT #sc-2003) were used instead of magnetic beads. For sonication, a Bioruptor Pico sonicator (Diagenode # B01060010) was used with mode set at 20 cycles 30 seconds on / off. For p53 immunoprecipitation DO-1 (SCBT #sc-126) and normal mouse IgG (SCBT #2025) were used with enrichment calculated as $2^{-(DO-1 \text{ Cq} - \text{IgG Cq})}$. qPCR reactions were performed using the Maxima SYBR Green qPCR Master Mix (Life Technologies) on a Bio-Rad CFX96 qPCR system. ChIP primer sequences: *CDKN1A* forward: 5'-AGCAGGCTGTGGCTCTGATT-3', *CDKN1A* reverse: 5'-CAAAATAGCCACCAGCCTCTTCT-3'; *MDM2* forward: 5'-CGTTCCGAAACTGCAGTAAA-

3', *MDM2* reverse: 5'-CAGCTGGAGACAAGTCAGGA-3'. Each sample was run in biological and technical duplicate.

Compounds

PhiKan083 (Sigma-Aldrich #SML1770), PK11000 (Enamine # EN300-27043), Nutlin-3a (Apexbio #A3671)

1-carbazol-9-ylprop-2-en-1-one (KG1). An oven dried reaction vessel was placed under argon and charged with 9H-carbazole (100 mg, 0.598 mmol) and sodium hydride (21.53 mg, 0.897 mmol). THF (2 mL) was added and the reaction was stirred at 4°C for 10 minutes. prop-2-enoyl prop-2-enoate (113.1 mg, 0.897 mmol) was added and the reaction was warmed from 4°C to room temperature over 30 minutes. The reaction mixture was quenched with saturated sodium bicarbonate solution and extracted in dichloromethane. The mixture was washed in brine solution, dried with sodium sulfate, and concentrated *in vacuo*. The crude was purified by flash chromatography over silica gel eluting with a gradient from 0-15% ethyl acetate-hexanes to afford KG1 (21 mg, 0.095 mmol, 15.87% yield) as a white semi-solid. ¹H NMR (400 MHz, DMSO-*d*₆) δ 8.22 (ddd, *J* = 7.6, 1.4, 0.7 Hz, 2H), 8.12 (dt, *J* = 8.4, 0.9 Hz, 2H), 7.54 (ddd, *J* = 8.5, 7.3, 1.4 Hz, 2H), 7.45 (td, *J* = 7.5, 1.0 Hz, 2H), 7.27 (dd, *J* = 16.9, 10.6 Hz, 1H), 6.55 (dd, *J* = 16.9, 1.5 Hz, 1H), 6.18 (dd, *J* = 10.5, 1.5 Hz, 1H). HRMS(ESI-TOF): *m/z*: calculated for C₁₆H₁₁NO₂ [M + H]⁺ 249.3, found 250.0866

1-carbazol-9-yl-2-chloro-ethanone (KG2). An oven dried reaction vessel was placed under argon and charged with 9H-carbazole (100 mg, 0.598 mmol) and sodium hydride (21.53 mg, 0.897 mmol). THF (2 mL) was added and the reaction was stirred at 4°C for 10 minutes. Chloroacetyl chloride (101.3, 0.897 mmol) was added and the reaction was warmed from 4°C to room

temperature over 30 minutes. The reaction mixture was quenched with saturated sodium bicarbonate solution and extracted in dichloromethane. The mixture was washed in brine solution, dried with sodium sulfate, and concentrated *in vacuo*. The crude was purified by flash chromatography over silica gel eluting with a gradient from 0-15% ethyl acetate-hexanes to afford KG2 (9.9 mg, 0.041 mmol, 6.793% yield) as a clear semi-solid. ¹H NMR (400 MHz, DMSO-*d*₆) δ 8.50 – 8.12 (m, 4H), 7.75 – 7.38 (m, 4H), 5.35 (s, 2H). HRMS(ESI-TOF): *m/z*: calculated for C₁₅H₁₁NO [M + H]⁺ 221.3, found 222.0939

9-prop-2-enoylcarbazole-3-carbaldehyde (KG3). An oven dried reaction vessel was placed under argon and charged with 9H-carbazole-3-carbaldehyde (100 mg, 0.512 mmol). dichloromethane (2 mL), triethylamine (214.19 uL, 1.537 mmol), and prop-2-enoyl prop-2-enoate (70.47 uL, 0.615 mmol) were added and the reaction was stirred at room temperature for 1 hr under argon. The reaction mixture was quenched with saturated sodium bicarbonate solution and extracted in dichloromethane. The mixture was washed in brine solution, dried with sodium sulfate, and concentrated *in vacuo*. The crude was purified by flash chromatography over silica gel eluting with a gradient from 0-50% ethyl acetate-hexanes to afford KG3 (25 mg, 0.1 mmol, 19.58% yield) as a yellow semisolid. ¹H NMR (400 MHz, DMSO-*d*₆) δ 10.15 (s, 1H), 8.81 (dd, *J* = 1.7, 0.6 Hz, 1H), 8.42 – 8.26 (m, 2H), 8.15 – 8.05 (m, 2H), 7.66 – 7.47 (m, 2H), 7.27 (dd, *J* = 16.9, 10.6 Hz, 1H), 6.60 (dd, *J* = 16.9, 1.5 Hz, 1H), 6.24 (dd, *J* = 10.6, 1.4 Hz, 1H). HRMS(ESI-TOF): *m/z*: calculated for C₁₄H₁₀ClNO [M + H]⁺ 243.7, found 244.0532.

9-prop-2-enoylcarbazole-3-carbaldehyde (KG4). An oven dried reaction vessel was placed under argon and charged with 9H-carbazole-3-carbaldehyde (100 mg, 0.512 mmol) and N-Methylpropargyl amine (106.2 mg, 1.537 mmol). Methanol (1.8 mL) and glacial acetic acid (0.2 mL) were added and the reaction was stirred at room temperature for 30 minutes. Sodium

cyanoborohydride (38.63 mg, 0.615 mmol) was added and the reaction proceeded overnight. The reaction mixture was quenched with saturated sodium bicarbonate solution and extracted in ethyl acetate. The mixture was washed in brine solution, dried with sodium sulfate, and concentrated *in vacuo*. The crude was purified by flash chromatography over silica gel eluting with a gradient from 0-100% ethyl acetate-hexanes to afford **N-(9H-carbazol-3-ylmethyl)-N-methyl-prop-2-yn-1-amine** (108 mg, 0.435 mmol, 84.91% yield) as a white/yellow semi-solid and was used in the next reaction step. An oven dried reaction vessel was placed under argon and charged with N-(9H-carbazol-3-ylmethyl)-N-methyl-prop-2-yn-1-amine (108 mg, 0.435 mmol) and sodium hydride (20.88 mg, 0.870 mmol). THF (2 mL) was added and the reaction was stirred at 4°C for 10 minutes. prop-2-enoyl prop-2-enoate (82.27 mg, 0.652 mmol) was added and the reaction was warmed from 4°C to room temperature over 30 minutes. The reaction mixture was quenched with saturated sodium bicarbonate solution and extracted in dichloromethane. The mixture was washed in brine solution, dried with sodium sulfate, and concentrated *in vacuo*. The crude was purified by flash chromatography over silica gel eluting with a gradient from 0-80% ethyl acetate-hexanes to afford KG4 (3.9 mg, 0.013 mmol, 2.966% yield) as a clear semi-solid. ¹H NMR (400 MHz, DMSO-*d*₆) δ 8.53 – 8.00 (m, 3H), 7.71 – 7.09 (m, 3H), 6.54 (dd, *J* = 16.9, 1.6 Hz, 1H), 6.18 (dd, *J* = 10.5, 1.6 Hz, 1H), 3.69 (s, 2H), 3.23 (t, *J* = 2.4 Hz, 1H), 2.26 (s, 2H), 1.24 (d, *J* = 3.1 Hz, 3H), 0.86 (t, *J* = 6.5 Hz, 1H). HRMS(ESI-TOF): *m/z*: calculated for C₂₀H₁₈N₂O [M + H]⁺ 302.4, found 235.0956 (loss of NCH₃CH₂CCH).

9-prop-2-enoyl-N-prop-2-ynyl-carbazole-3-carboxamide (KG5). An oven dried reaction vessel was placed under argon and charged with 9H-carbazole-3-carboxylic acid (200 mg, 0.947 mmol) and HATU (0.947 mg, 1.42 mmol). DMF (2 mL) and N,N-diisopropylpropan-2-amine (407 mg, 2.841 mmol) were added and the reaction was stirred at room temperature for 10 minutes under

argon. Prop-2-yn-1-amine (104.3 mg, 1.894 mmol) was added and the reaction proceeded overnight. The product was extracted in ethyl acetate, washed in 5% citric acid solution, brine solution, sodium bicarbonate solution, brine solution, dried with sodium sulfate, and concentrated *in vacuo*. The crude was purified by flash chromatography over silica gel eluting with a gradient from 0-100% ethyl acetate-hexanes to afford **N-prop-2-ynyl-9H-carbazole-3-carboxamide** (225 mg, 0.906 mmol, 95.71% yield) as a white/yellow semi-solid and was used in the next reaction step. An oven dried reaction vessel was placed under argon and charged with N-prop-2-ynyl-9H-carbazole-3-carboxamide (100 mg, 0.403 mmol). dichloromethane (2 mL), triethylamine (122.3 mg, 1.208 mmol), and prop-2-enoyl prop-2-enoate (60.95 mg, 0.483 mmol) were added and the reaction was stirred at room temperature for 1 hr under argon. The reaction mixture was quenched with saturated sodium bicarbonate solution and extracted in dichloromethane. The mixture was washed in brine solution, dried with sodium sulfate, and concentrated *in vacuo*. The crude was purified by flash chromatography over silica gel eluting with a gradient from 0-50% ethyl acetate-hexanes to afford KG5 (72 mg, 0.238 mmol, 59.13 % yield) as a yellow semisolid.

¹H NMR (400 MHz, DMSO-*d*₆) δ 9.07 (t, *J* = 5.5 Hz, 1H), 8.74 (d, *J* = 1.7 Hz, 1H), 8.28 – 8.00 (m, 4H), 7.63 – 7.43 (m, 2H), 7.27 (dd, *J* = 16.9, 10.5 Hz, 1H), 6.58 (dd, *J* = 17.0, 1.5 Hz, 1H), 6.22 (dd, *J* = 10.5, 1.5 Hz, 1H), 4.14 (dd, *J* = 5.6, 2.5 Hz, 2H), 3.17 (t, *J* = 2.5 Hz, 1H).

HRMS(ESI-TOF): *m/z*: calculated for C₁₉H₁₄N₂O₂ [M + H]⁺ 302.3, found 303.116.

1-(2-methylprop-2-enoyl)indole-3-carbaldehyde (KG6). An oven dried reaction vessel was placed under argon and charged with 1H-indole-3-carbaldehyde (50 mg, 0.344mmol). Dichloromethane (2 mL), triethylamine (104.6 mg, 1.033 mmol), and methacrylic anhydride (63.72 mg, 0.413 mmol) were added and the reaction was stirred at room temperature 1 hr under argon. The reaction mixture was quenched with saturated sodium bicarbonate solution and

extracted in dichloromethane. The mixture was washed in brine solution, dried with sodium sulfate, and concentrated *in vacuo*. The crude was purified by flash chromatography over silica gel eluting with a gradient from 0-50% ethyl acetate-hexanes to afford KG6 (55.4 mg, 0.260 mmol, 75.44% yield) as a white semi-solid. ¹H NMR (400 MHz, Chloroform-*d*) δ 10.18 (s, 1H), 8.48 (dt, *J* = 8.4, 1.0 Hz, 1H), 8.35 – 8.29 (m, 1H), 8.20 (s, 1H), 7.55 – 7.43 (m, 2H), 7.04 (dd, *J* = 16.8, 10.4 Hz, 1H), 6.80 (dd, *J* = 16.7, 1.2 Hz, 1H), 6.22 (dd, *J* = 10.5, 1.2 Hz, 1H), 1.28 (s, 1H), 0.88 (d, *J* = 17.5 Hz, 1H). HRMS(ESI-TOF): *m/z*: calculated for C₁₃H₁₁NO₂ [M + H]⁺ 213.2, found 214.0885.

4-methyl-1-(2-methylprop-2-enoyl)indole-3-carbaldehyde (KG7). An oven dried reaction vessel was placed under argon and charged with 4-methyl-1H-indole-3-carbaldehyde (100 mg, 0.628 mmol). Dichloromethane (2 mL), triethylamine (190.7 mg, 1.885 mmol), and methacrylic anhydride (116.2 mg, 0.754 mmol) were added and the reaction was stirred at room temperature 1 hr under argon. The reaction mixture was quenched with saturated sodium bicarbonate solution and extracted in dichloromethane. The mixture was washed in brine solution, dried with sodium sulfate, and concentrated *in vacuo*. The crude was purified by flash chromatography over silica gel eluting with a gradient from 0-50% ethyl acetate-hexanes to afford KG7 (73.4 mg, 0.323mmol, 51.41% yield) as a white semi-solid. ¹H NMR (400 MHz, DMSO-*d*₆) δ 10.06 (s, 1H), 8.58 (s, 1H), 8.13 (dt, *J* = 8.3, 0.9 Hz, 1H), 7.36 (dd, *J* = 8.3, 7.4 Hz, 1H), 7.22 (dt, *J* = 7.4, 1.0 Hz, 1H), 5.97 (q, *J* = 1.6 Hz, 1H), 5.68 (d, *J* = 1.2 Hz, 1H), 2.78 (s, 3H), 2.14 (dd, *J* = 1.6, 1.0 Hz, 3H). HRMS(ESI-TOF): *m/z*: calculated for C₁₄H₁₃NO₂ [M + H]⁺ 227.3, found 228.1046.

1-(2-methylprop-2-enoyl)-4-pyrimidin-5-yl-indole-3-carbaldehyde (KG8). An oven dried reaction vessel was placed under argon and charged with 4-bromo-1H-indole-3-carbaldehyde (100 mg, 0.446 mmol), cesium carbonate (290.8 mg, 0.893 mmol), 1,1'-Bis(di-tert-

butylphosphino)ferrocene-palladium dichloride (14.63 mg, 0.050 mmol), and pyrimidin-5-ylboronic acid (82.96 mg, 0.669 mmol). Water (1 mL) and acetonitrile (1 mL) were degassed with argon and were added to the reaction vessel. The reaction was stirred at 80°C for 24 hrs, cooled to room temperature, and extracted in ethyl acetate, washed in brine solution, dried with sodium sulfate, and concentrated *in vacuo*. The crude was purified by flash chromatography over silica gel eluting with a gradient from 0-100% ethyl acetate-hexanes to afford 4-pyrimidin-5-yl-1H-indole-3-carbaldehyde (20 mg, 0.090 mmol, 20.07% yield) as a brown/orange solid and was used in the next reaction step. An oven dried reaction vessel was placed under argon and charged with 4-pyrimidin-5-yl-1H-indole-3-carbaldehyde (20 mg, 0.090 mmol). Dichloromethane (2 mL), triethylamine (27.2 mg, 0.269 mmol), and Methacrylic anhydride (16.57 mg, 0.108mmol) were added and the reaction was stirred at room temperature for 1 hr under argon. The reaction mixture was quenched with saturated sodium bicarbonate solution and extracted in dichloromethane. The mixture was washed in brine solution, dried with sodium sulfate, and concentrated *in vacuo*. The crude was purified by flash chromatography over silica gel eluting with a gradient from 0-50% ethyl acetate-hexanes to afford KG8 (12 mg, 0.8024 mmol, 78% yield) as a yellow semisolid. ¹H NMR (400 MHz, DMSO-*d*₆) δ 9.77 (s, 1H), 9.21 (s, 1H), 8.78 (d, *J* = 9.0 Hz, 3H), 8.43 (dd, *J* = 8.4, 1.0 Hz, 1H), 7.69 – 7.60 (m, 1H), 7.46 (dd, *J* = 7.5, 1.0 Hz, 1H), 6.03 (q, *J* = 1.6 Hz, 1H), 2.17 (d, *J* = 1.3 Hz, 3H), 1.84 (t, *J* = 1.3 Hz, 1H). HRMS(ESI-TOF): *m/z*: calculated for C₁₇H₁₃N₃O₂ [M + H]⁺ 291.3, found 292.1141.

4-[2-(4-methylpiperazin-1-yl)pyrimidin-5-yl]-1-(2-methylprop-2-enoyl)indole-3-carbaldehyde (KG9). An oven dried reaction vessel was placed under argon and charged with 4-bromo-1H-indole-3-carbaldehyde (400 mg, 1.785 mmol), cesium carbonate (1163 mg, 3.571 mmol), 1,1'-Bis(di-tert-butylphosphino)ferrocene-palladium dichloride (58.54 mg, 0.089 mmol), and 2-(4-

methylpiperazin-1-yl)-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyrimidine (814.6 mg, 2.678 mmol). Water (2 mL) and acetonitrile (2 mL) were degassed with argon and were added to the reaction vessel. The reaction was stirred at 80°C for 24 hrs, cooled to room temperature, and extracted in ethyl acetate, washed in brine solution, dried with sodium sulfate, and concentrated *in vacuo*. The crude was purified by flash chromatography over silica gel eluting with a gradient from 0-20% methanol-dichloromethane to afford 4-[2-(4-methylpiperazin-1-yl)pyrimidin-5-yl]-1H-indole-3-carbaldehyde (443.9 mg, 1.381 mmol, 77.37 % yield) as a brown/red solid and was used in the next reaction step. An oven dried reaction vessel was placed under argon and charged with 4-[2-(4-methylpiperazin-1-yl)pyrimidin-5-yl]-1H-indole-3-carbaldehyde (100 mg, 0.311 mmol). Dichloromethane (2 mL), triethylamine (94.46 mg, 0.933 mmol), and methacrylic anhydride (57.56 mg, 0.373 mmol) were added and the reaction was stirred at room temperature for 1 hr under argon. The reaction mixture was quenched with saturated sodium bicarbonate solution and extracted in dichloromethane. The mixture was washed in brine solution, dried with sodium sulfate, and concentrated *in vacuo*. The crude was purified by flash chromatography over silica gel eluting with a gradient from 0-20% methanol-dichloromethane to afford KG9 (65 mg, 0.167 mmol, 53.64 % yield) as a brown semi-solid. ¹H NMR (400 MHz, DMSO-*d*₆) δ 12.44 (s, 1H), 9.70 (s, 1H), 8.36 (d, *J* = 2.0 Hz, 3H), 7.55 (d, *J* = 8.0 Hz, 1H), 7.35 (t, *J* = 7.7 Hz, 1H), 7.12 (d, *J* = 7.3 Hz, 1H), 5.97 (s, 1H), 5.60 (t, *J* = 1.9 Hz, 1H), 3.80 (t, *J* = 4.9 Hz, 4H), 2.41 (t, *J* = 5.0 Hz, 4H), 2.25 (s, 3H), 1.85 (s, 2H). HRMS(ESI-TOF): *m/z*: calculated for C₂₂H₂₄N₅O₂⁺ [M + H]⁺ 390.19, found 390.2006.

4-[4-(4-methylpiperazin-1-yl)phenyl]-1-(2-methylprop-2-enoyl)indole-3-carbaldehyde (KG10).

An oven dried reaction vessel was placed under argon and charged with 4-bromo-1H-indole-3-carbaldehyde (400 mg, 1.785 mmol), cesium carbonate (1163 mg, 3.571 mmol), 1,1'-Bis(di-tert-

butylphosphino)ferrocene-palladium dichloride (58.54 mg, 0.089 mmol), and 1-methyl-4-[4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)phenyl]piperazine (809.3 mg, 2.678 mmol). Water (2 mL) and acetonitrile (2 mL) were degassed with argon and were added to the reaction vessel. The reaction was stirred at 80°C for 24 hrs, cooled to room temperature, and extracted in ethyl acetate, washed in brine solution, dried with sodium sulfate, and concentrated *in vacuo*. The crude was purified by flash chromatography over silica gel eluting with a gradient from 0-20% methanol-dichloromethane to afford 4-[4-(4-methylpiperazin-1-yl)phenyl]-1H-indole-3-carbaldehyde (266 mg, 0.833 mmol, 46.65 % yield) as a brown/red solid and was used in the next reaction step. An oven dried reaction vessel was placed under argon and charged with 4-[4-(4-methylpiperazin-1-yl)phenyl]-1H-indole-3-carbaldehyde (100 mg, 0.313 mmol).

Dichloromethane (2 mL), triethylamine (95.04 mg, 0.939 mmol), and methacrylic anhydride (57.92 mg, 0.376 mmol) were added and the reaction was stirred at room temperature for 1 hr under argon. The reaction mixture was quenched with saturated sodium bicarbonate solution and extracted in dichloromethane. The mixture was washed in brine solution, dried with sodium sulfate, and concentrated *in vacuo*. The crude was purified by flash chromatography over silica gel eluting with a gradient from 0-20% methanol-dichloromethane to afford KG10 (76 mg, 0.196 mmol, 62.65 % yield) as a brown semi-solid. ¹H NMR (400 MHz, DMSO-*d*₆) δ 8.34 – 8.22 (m, 1H), 7.58 – 7.27 (m, 5H), 7.12 – 6.97 (m, 3H), 6.04 – 5.93 (m, 1H), 5.60 (q, *J* = 1.7 Hz, 1H), 3.22 (q, *J* = 5.2 Hz, 6H), 2.25 (s, 3H), 2.12 (s, 1H), 1.85 (t, *J* = 1.2 Hz, 2H). HRMS(ESI-TOF): *m/z*: calculated for C₂₄H₂₇N₃O₂²⁺ [M + H]⁺ 389.21, found 389.2105

2-methyl-1-[3-methyl-4-[4-(4-methylpiperazin-1-yl)phenyl]pyrrolo[2,3-*c*]pyridin-1-yl]prop-2-en-1-one (KG11). An oven dried reaction vessel was placed under argon and charged with 4-bromo-3-methyl-1H-pyrrolo[2,3-*c*]pyridine (200 mg, 0.948 mmol), cesium carbonate (617.5 mg,

1.895 mmol), 1,1'-Bis(di-tert-butylphosphino)ferrocene-palladium dichloride (31.07 mg, 0.047 mmol), and 1-methyl-4-[4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)phenyl]piperazine (429.6 mg, 1.421 mmol). Water (2 mL) and acetonitrile (2 mL) were degassed with argon and were added to the reaction vessel. The reaction was stirred at 80°C for 24 hrs, cooled to room temperature, and extracted in ethyl acetate, washed in brine solution, dried with sodium sulfate, and concentrated *in vacuo*. The crude was purified by flash chromatography over silica gel eluting with a gradient from 0-20% methanol-dichloromethane to afford 3-methyl-4-[4-(4-methylpiperazin-1-yl)phenyl]-1H-pyrrolo[2,3-c]pyridine (127 mg, 0.414 mmol, 43.69% yield) as a white/green solid and was used in the next reaction step. An oven dried reaction vessel was placed under argon and charged with 3-methyl-4-[4-(4-methylpiperazin-1-yl)phenyl]-1H-pyrrolo[2,3-c]pyridine (127 mg, 0.414 mmol) and Sodium hydride (19.9 mg, 0.829 mmol). DMF (2 mL) was added and the reaction was stirred at 4°C for 10 minutes under argon. Methacrylic anhydride (69.71 uL, 0.497 mmol) was added at 4°C the reaction mixture was warmed to room temperature over 1 hr. The reaction mixture was quenched with saturated sodium bicarbonate solution and extracted in dichloromethane, dried with sodium sulfate, and concentrated *in vacuo*. The crude was purified by HPLC eluting with a gradient from 5-50% water/acetonitrile with 0.1% formic acid. HPLC fractions were pooled and the pH was adjusted with saturated sodium bicarbonate solution and extracted in ethyl acetate, dried with sodium sulfate, and concentrated *in vacuo* to afford KG11 (85 mg, 0.0227 mmol, 54.76 % yield) as a yellow solid. ¹H NMR (400 MHz, DMSO-*d*₆) δ 9.46 (s, 1H), 8.21 (s, 1H), 7.70 (d, *J* = 1.4 Hz, 1H), 7.32 – 7.26 (m, 2H), 7.08 – 7.01 (m, 2H), 5.84 (q, *J* = 1.6 Hz, 1H), 5.57 (d, *J* = 1.2 Hz, 1H), 3.22 (dd, *J* = 6.3, 3.8 Hz, 4H), 2.48 (d, *J* = 5.0 Hz, 5H), 2.25 (s, 3H), 2.12 (d, *J* = 1.3 Hz, 3H), 1.87 (d, *J* = 1.3 Hz, 3H). HRMS(ESI-TOF): *m/z*: calculated for C₂₃H₂₇N₄O⁺ [*M* + *H*]⁺ 375.22, found 375.2213

2-methyl-1-[3-methyl-4-[4-(1-methyl-4-piperidyl)phenyl]pyrrolo[2,3-c]pyridin-1-yl]prop-2-en-1-

one (KG12). An oven dried reaction vessel was placed under argon and charged with 4-bromo-3-methyl-1H-pyrrolo[2,3-c]pyridine (200 mg, 0.948 mmol), cesium carbonate (617.5 mg, 1.895 mmol), 1,1'-Bis(di-tert-butylphosphino)ferrocene-palladium dichloride (31.07 mg, 0.047 mmol), and 1-methyl-4-[4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)phenyl]piperidine (285.4 mg, 0.948 mmol). Water (2 mL) and acetonitrile (2 mL) were degassed with argon and were added to the reaction vessel. The reaction was stirred at 80°C for 24 hrs, cooled to room temperature, and extracted in ethyl acetate, washed in brine solution, dried with sodium sulfate, and concentrated *in vacuo*. The crude was purified by flash chromatography over silica gel eluting with a gradient from 0-20% methanol-dichloromethane to afford 3-methyl-4-[4-(1-methyl-4-piperidyl)phenyl]-1H-pyrrolo[2,3-c]pyridine (133.1 mg, 0.436 mmol, 45.99 % yield) as a white/brown semi-solid and was used in the next reaction step. An oven dried reaction vessel was placed under argon and charged with 3-methyl-4-[4-(1-methyl-4-piperidyl)phenyl]-1H-pyrrolo[2,3-c]pyridine (133.1 mg, 0.436 mmol) and Sodium hydride (20.92 mg, 0.872 mmol). DMF (2 mL) was added and the reaction was stirred at 4°C for 10 minutes under argon. Methacrylic anhydride (73.29 uL, 0.523 mmol) was added at 4°C and the reaction mixture was warmed to room temperature over 1 hr. The reaction mixture was quenched with saturated sodium bicarbonate solution and extracted in dichloromethane, dried with sodium sulfate, and concentrated *in vacuo*. The crude was purified by HPLC eluting with a gradient from 5-50% water/acetonitrile with 0.1% formic acid. HPLC fractions were pooled and the pH was adjusted with saturated sodium bicarbonate solution and extracted in ethyl acetate, dried with sodium sulfate, and concentrated *in vacuo* to afford KG12 (71 mg, 0.0190 mmol, 43.62 % yield) as a white solid. ¹H NMR (400 MHz, DMSO-*d*₆) δ 9.49 (s, 1H), 8.24 (s, 1H), 7.72 (t, *J* = 1.3 Hz, 1H), 7.38 (s, 4H), 5.84 (t, *J* = 1.7 Hz, 1H), 5.57 (s, 1H), 2.90

(dt, $J = 12.0, 3.2$ Hz, 2H), 2.30 – 2.06 (m, 7H), 2.00 (td, $J = 11.5, 2.6$ Hz, 2H), 1.91 – 1.62 (m, 8H). HRMS(ESI-TOF): m/z : calculated for $C_{24}H_{28}N_3O^+$ $[M + H]^+$ 374.22, found 374.2265.

2-methyl-1-[3-methyl-4-(2-methyl-3,4-dihydro-1H-isoquinolin-6-yl)pyrrolo[2,3-c]pyridin-1-yl]prop-2-en-1-one (KG13). An oven dried reaction vessel was placed under argon and charged

5 with 4-bromo-3-methyl-1H-pyrrolo[2,3-c]pyridine (200 mg, 0.948 mmol), cesium carbonate (617.5 mg, 1.895 mmol), 1,1'-Bis(di-tert-butylphosphino)ferrocene-palladium dichloride (31.07 mg, 0.047 mmol), and 2-methyl-6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-3,4-dihydro-1H-isoquinoline (388.3 mg, 1.421 mmol). Water (2 mL) and acetonitrile (2 mL) were degassed with argon and were added to the reaction vessel. The reaction was stirred at 80°C for 24 hrs, cooled to
10 room temperature, and extracted in ethyl acetate, washed in brine solution, dried with sodium sulfate, and concentrated *in vacuo*. The crude was purified by flash chromatography over silica gel eluting with a gradient from 0-20% methanol-dichloromethane to afford 2-methyl-6-(3-methyl-1H-pyrrolo[2,3-c]pyridin-4-yl)-3,4-dihydro-1H-isoquinoline (162.6 mg, 0.586 mmol, 61.87 % yield) as a brown semi-solid and was used in the next reaction step. An oven dried
15 reaction vessel was placed under argon and charged with 2-methyl-6-(3-methyl-1H-pyrrolo[2,3-c]pyridin-4-yl)-3,4-dihydro-1H-isoquinoline (162.6 mg, 0.586 mmol) and Sodium hydride (28.14 mg, 1.172 mmol). DMF (2 mL) was added and the reaction was stirred at 4°C for 10 minutes under argon. Methacrylic anhydride (98.59 μ L, 0.703 mmol) was added at 4°C and the reaction mixture was warmed to room temperature over 1 hr. The reaction mixture was quenched with
20 saturated sodium bicarbonate solution and extracted in dichloromethane, dried with sodium sulfate, and concentrated *in vacuo*. The crude was purified by HPLC eluting with a gradient from 5-50% water/acetonitrile with 0.1% formic acid. HPLC fractions were pooled and the pH was adjusted with saturated sodium bicarbonate solution and extracted in ethyl acetate, dried with

sodium sulfate, and concentrated *in vacuo* to afford KG13 (71 mg, 0.0190 mmol, 43.62 %
 yield) as a white solid. ¹H NMR (400 MHz, DMSO-*d*₆) δ 9.49 (s, 1H), 8.23 (s, 1H), 7.72 (d, *J* =
 1.4 Hz, 1H), 7.27 – 7.12 (m, 3H), 5.85 (t, *J* = 1.7 Hz, 1H), 5.57 (s, 1H), 3.56 (s, 2H), 2.90 (t, *J* =
 5.9 Hz, 2H), 2.64 (t, *J* = 5.9 Hz, 2H), 2.38 (s, 3H), 2.12 (d, *J* = 1.3 Hz, 3H), 1.83 (d, *J* = 1.3 Hz,
 3H). HRMS(ESI-TOF): *m/z*: calculated for C₂₂H₂₄N₃O⁺ [M + H]⁺ 346.19, found 346.1946.

Data Availability:

The crystal structures described in this study can be accessed at <https://www.rcsb.org/> under
 accession codes 8DC4 (KG3), 8DC6 (KG6), 8DC7 (KG10), and 8DC8 (KG13).

Acknowledgements:

Funding: This research was supported by grants from the Emerald Foundation, Inc., Samuel
 Waxman Cancer Research Foundation, and Howard Hughes Medical Institute to K.M.S. K.Z.G. is
 supported by a postdoctoral fellowship from Damon Runyon Research Foundation (DRG-2399-
 20). Data collection at the ALS Beamline 8.2.1 is supported by Howard Hughes Medical Institute.

Author Contributions: K.M.S. and K.Z.G. designed the studies and wrote the manuscript. K.Z.G.
 performed the experiments and experimental analysis.

We would like to thank Prof. Laura D. Attardi, Mengxiong Wang, Chad Altobelli, and Grant
 Burley for their insight on the study.

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Figures Legends:

Figure. 1: Development of a covalent PhiKan083 (A) Chemical structures of the carbazole series. (B) 1 μ M p53 Y220C-CL was incubated with 10 μ M of the carbazole compounds at 4°C for 24 hr. Adduct formation was observed by LC/MS. (C) HepG2 (p53 WT/+) or Huh-7 (p53 Y220C/-) cells were treated with the indicated KG5 concentration for 1 hr at 37°C and target engagement was observed by gel shift following a copper click reaction with TAMRA-azide. (D)

Crystal structure of p53 Y220C-CL bound to KG3 at 2.4 Å resolution. (E) Crystal structure of the p53 Y220C-PhiKan083 complex PDB 2VUK (8).

Figure. 2: A methacrylic indole compound induced a novel p53 conformational state (A)

Chemical structures of methacrylic indole compound series. (B) 1 μM p53 Y220C-CL was incubated with 10 μM indole series at 4°C for 24 hr. Adduct formation was analyzed by LC/MS. (C) Crystal structure of p53 Y220C-CL bound to KG6 in a novel conformational state at 1.6 Å resolution. (D) Crystal structure of p53 Y220C-CL bound to KG10 at 2.0 Å resolution.

Figure. 3: Azaindole compounds stabilized p53 Y220C to WT levels (A)

Chemical structures of the azaindole series. (B) p53 Y220C-CL was fully labeled by the compounds, purified by size-exclusion chromatography, and its thermal stability was analyzed by differential scanning fluorimetry. The difference in T_m relative to apo p53 Y220C-CL is displayed. (C) Crystal structure of p53 Y220C-CL bound to KG13 at 1.7 Å resolution. (D) 1 μM of p53 Y220C-CL or p53 Y220C-CL D228A were incubated with 10 μM compound at 4°C for 24 hr. Adduct formation was analyzed by LC/MS.

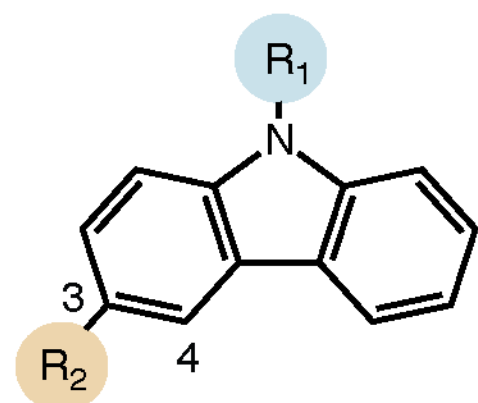
Figure. 4: KG13 stabilized cellular p53 Y220C and activated target gene expression (A)

1 μM of p53 WT or p53 Y220C were incubated with 10 μM of compound at 4°C for 24 hr. Adduct formation was analyzed by LC/MS. (B) Cellular thermal shift assay intensity values were plotted with calculated Tagg values where the KG13 treated sample showed an increase in thermal stability of cellular p53 Y220C compared to DMSO control. (C) RT-qPCR results for 25 μM KG13 treated H1299 cells stably expressing p53 R273C or Y220C. (D) RT-qPCR results for 10 μM Nutlin-3a or 25 μM KG13 treated U2OS cells stably expressing p53 Y220C. (E) Western blot for DMSO, 10 μM Nutlin-3a, and 10 μM KG13 treated cell panel. (F) Viability and caspase 3/7

activity for NUGC-4, NUGC-3, or BxPC-3 cells treated with KG13.

Figure 1

A



Compound

PK083

KG1

KG2

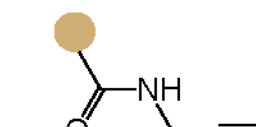
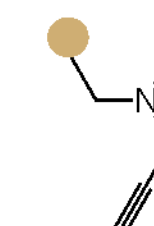
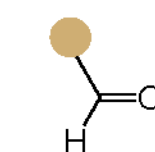
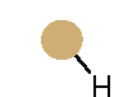
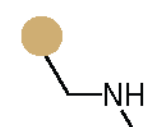
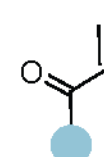
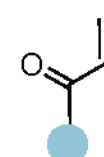
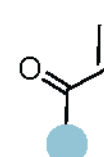
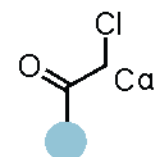
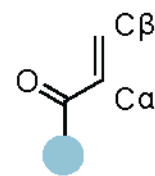
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KG4

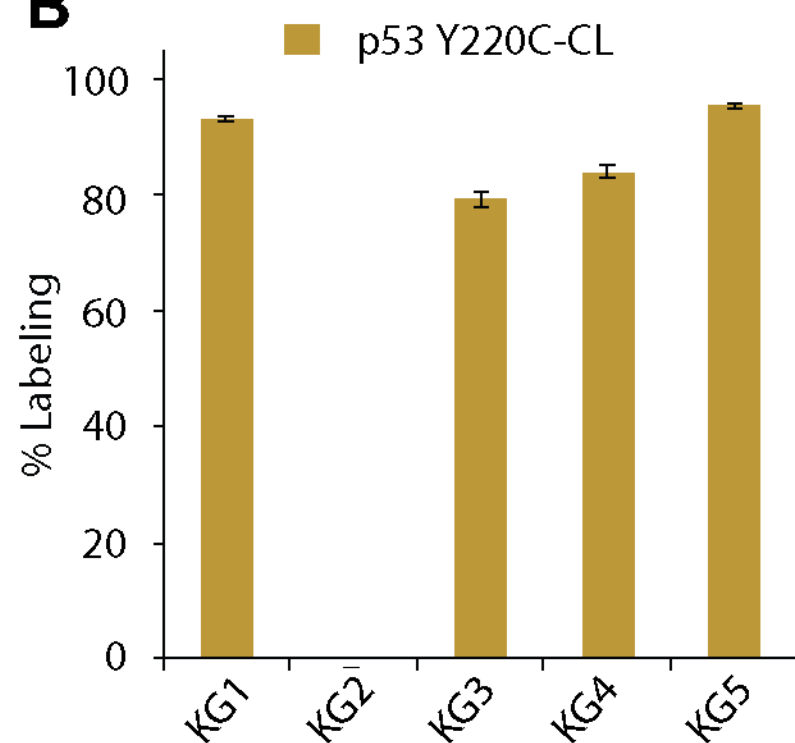
KG5

R₁

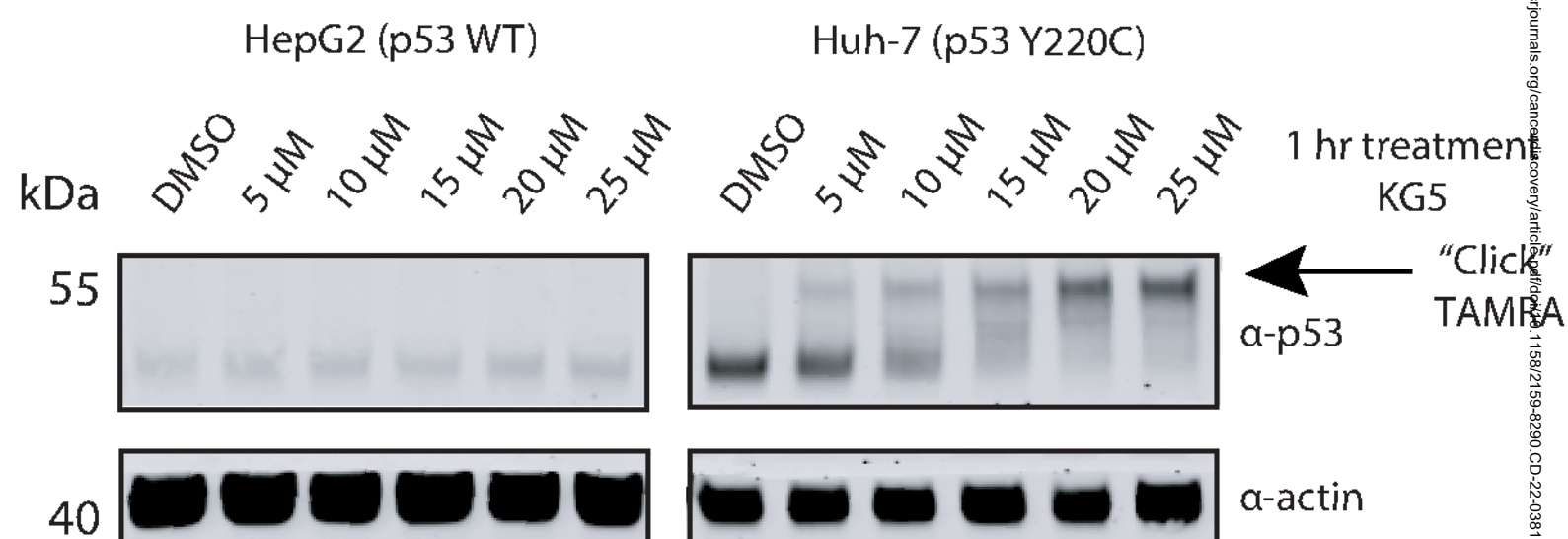
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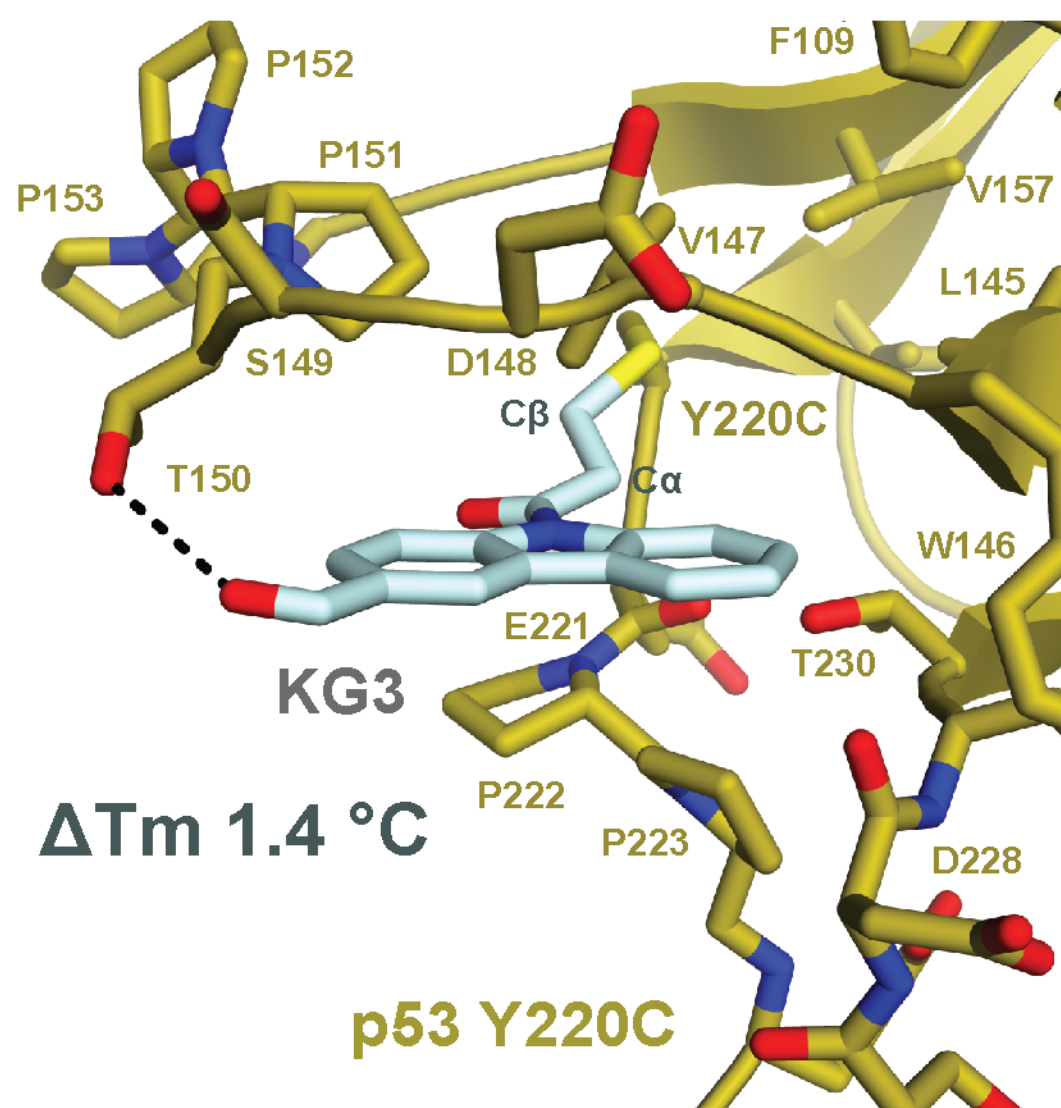
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C



D



E

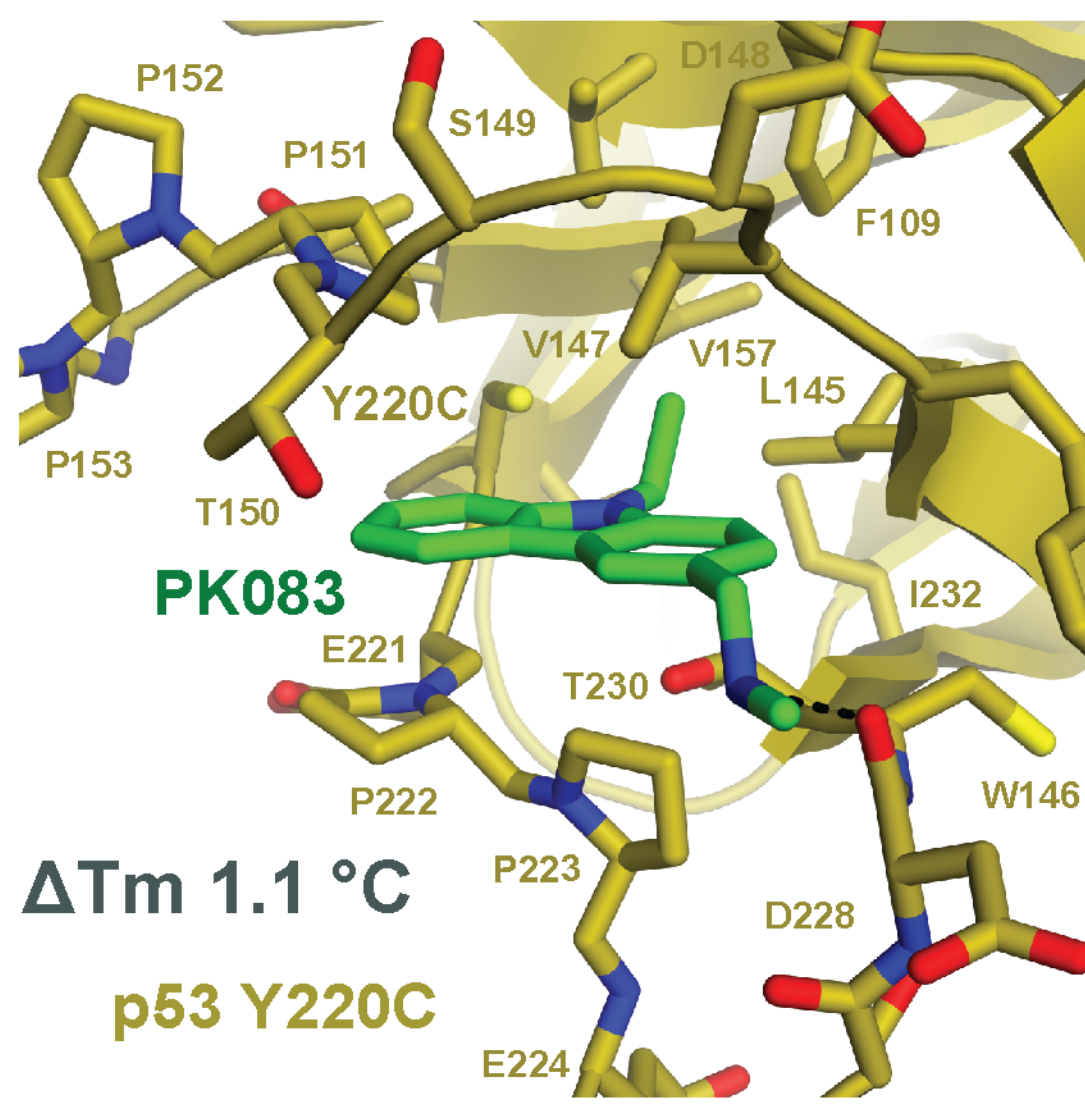


Figure 2

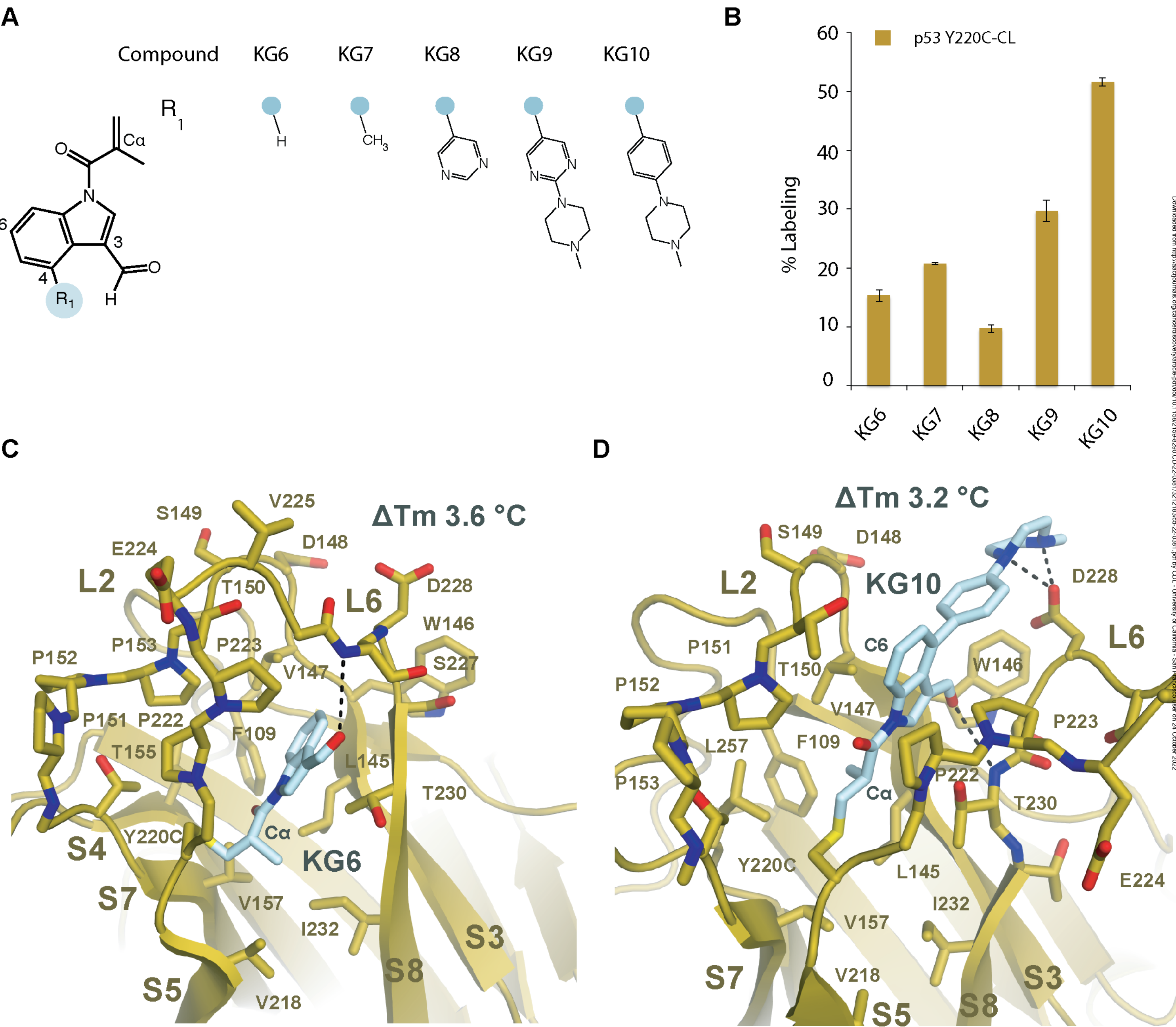
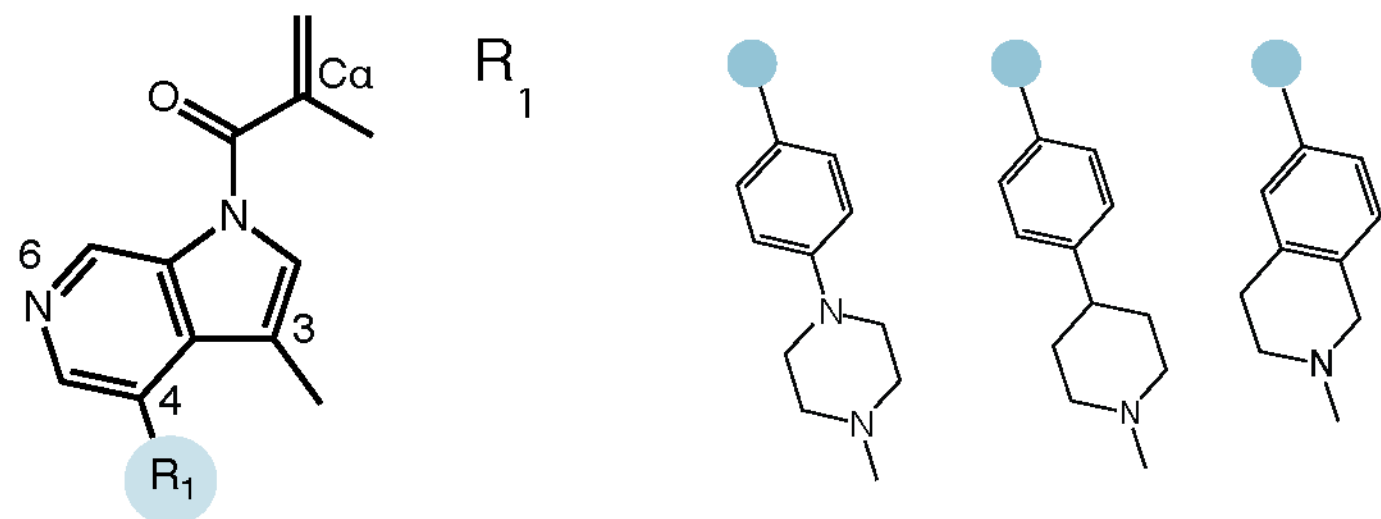
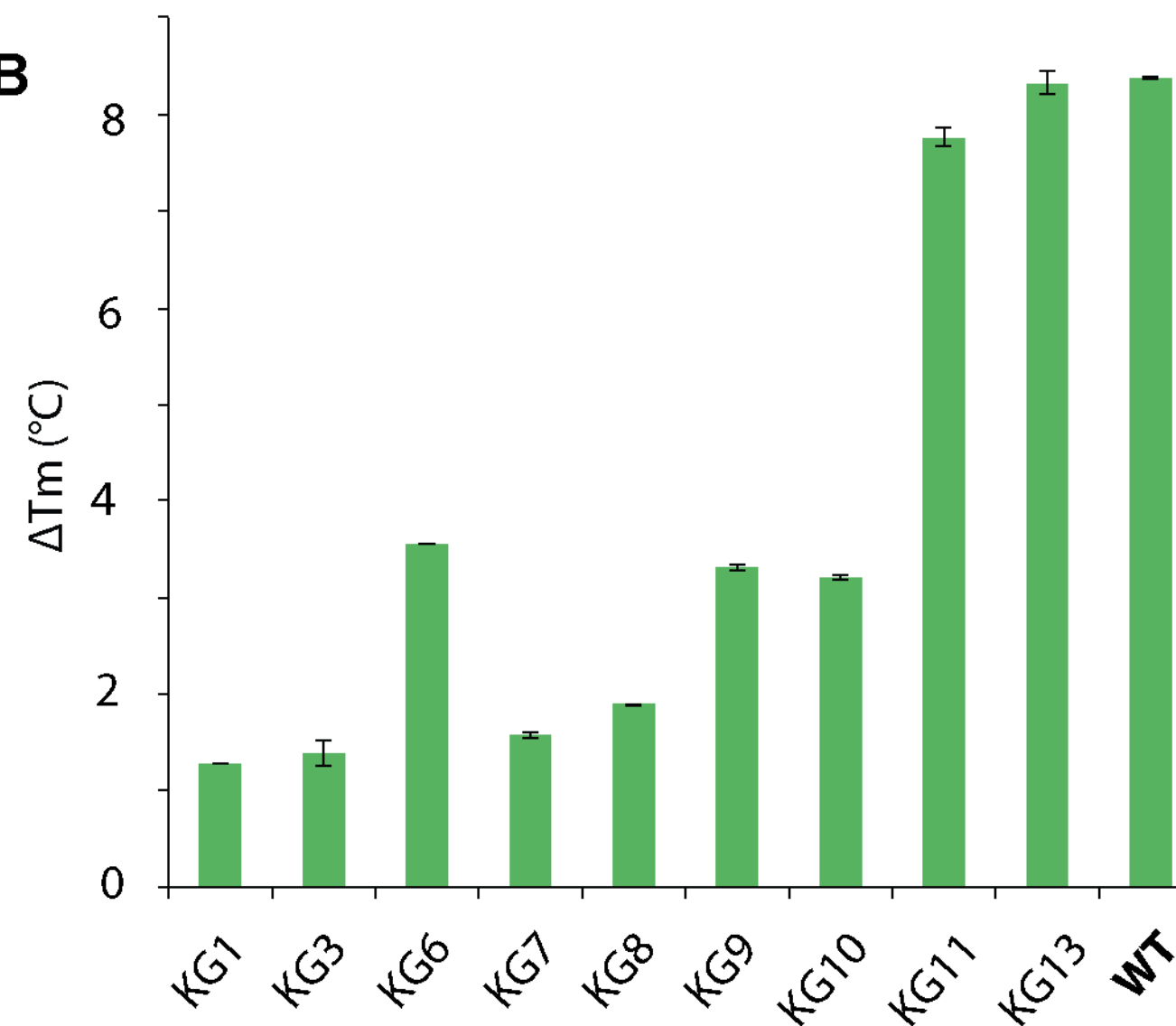


Figure 3

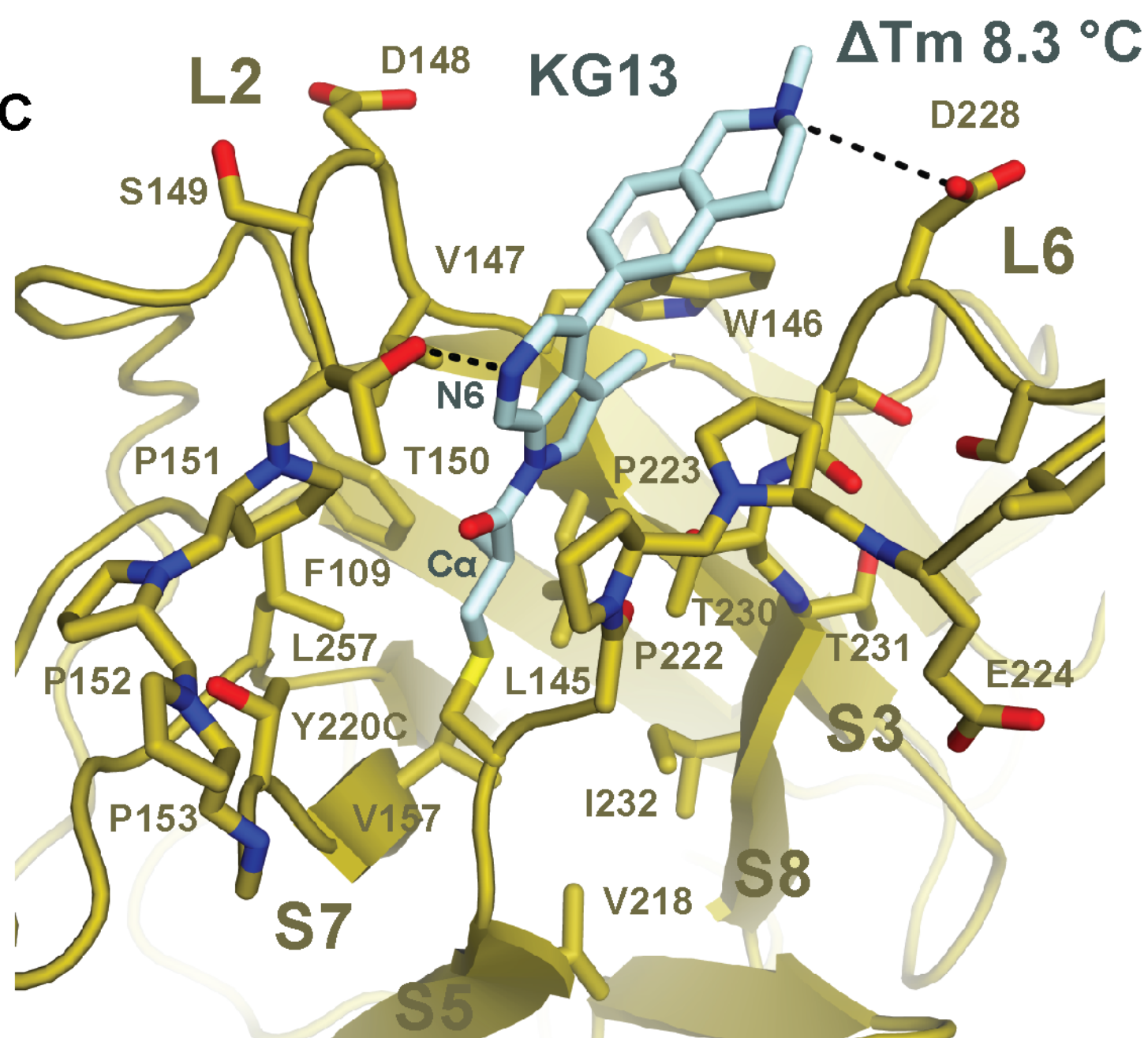
A



B



C



D

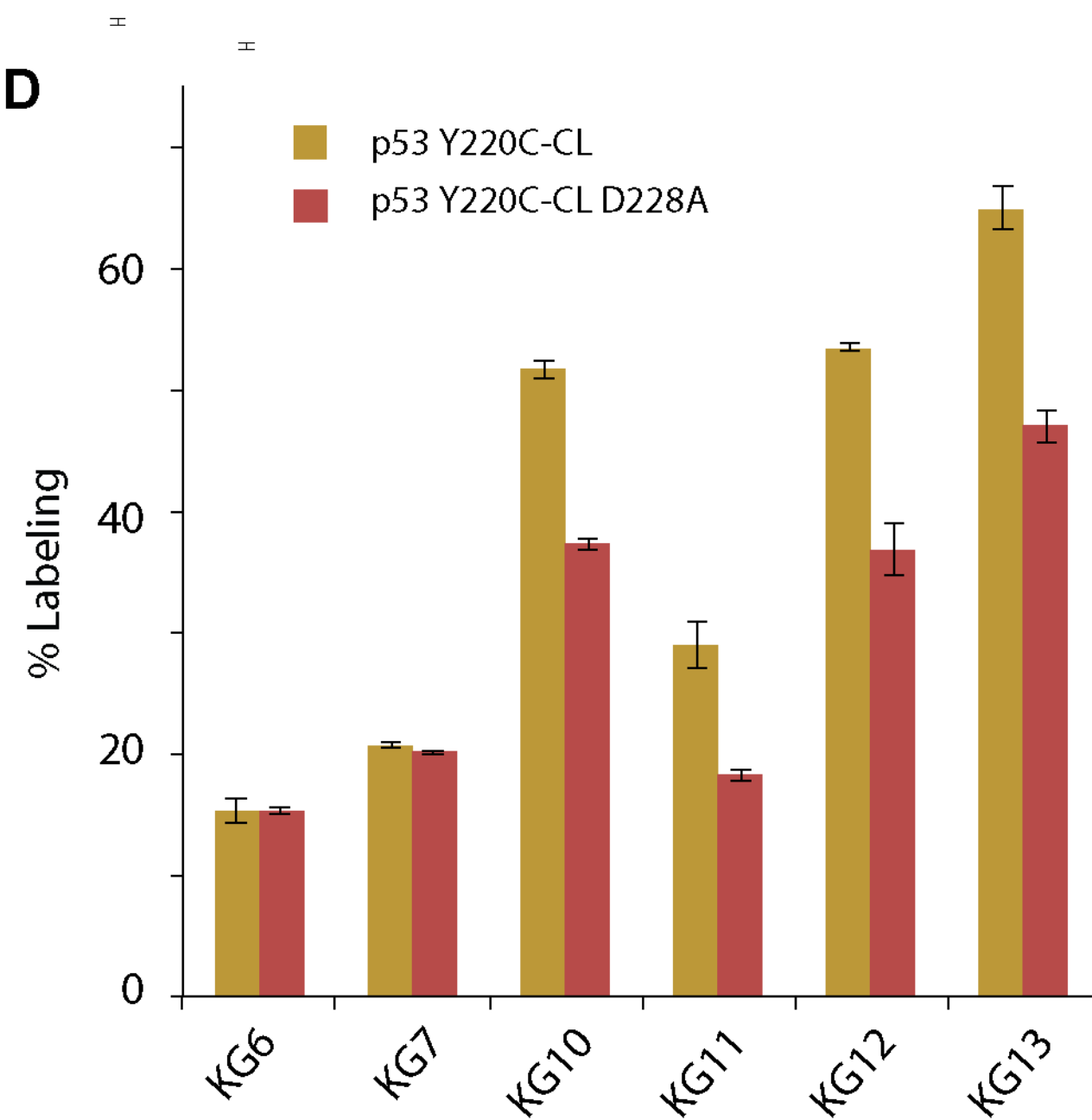


Figure 4

